Netweb TECHNOLOGIES

Tyrone A Netweb Brand

Empowering Compute, Network, Storage & AI

NVMe SSD Validation Report

4x Samsung PM9D3a - Cross-Server Comparison vs 172.16.13.217

Server / Host	user-Dev (172.16.15.38)
Platform	Tyrone MDI300 / 2x Intel Xeon 6730P (64C/128T), 1.0 TB DDR5
Validation Tier	Qualification
Campaign Window	04 Jun 2026 06:15-06:39 UTC
Prepared For	Netweb Technologies India Limited
Date	2026-06-04

Overall Result: PASS WITH LIMITATIONS

Executive Summary

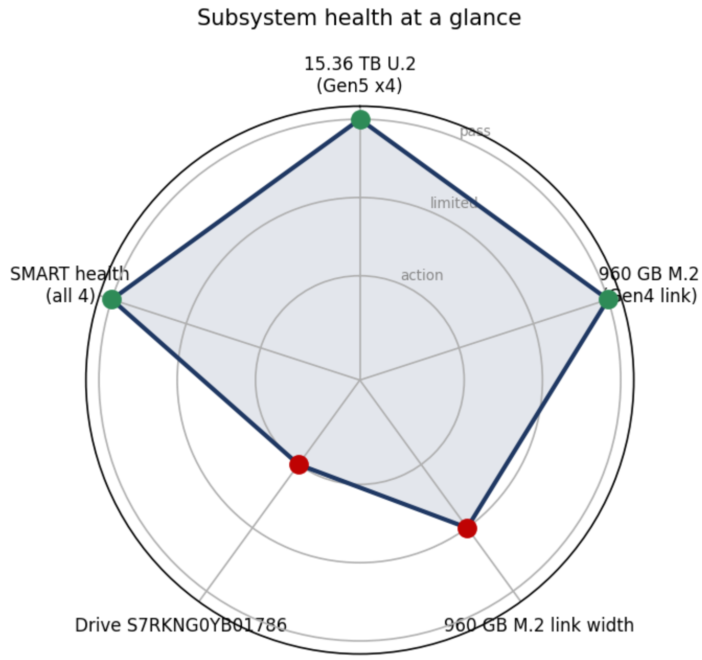
Re-validation of the four Samsung PM9D3a NVMe SSDs previously tested on the MDA200A2N-224 at 172.16.13.217, now installed in server 172.16.15.38 (Tyrone MDI300, hostname user-Dev; 2x Intel Xeon 6730P, 64C/128T, 1.0 TB DDR5). Serial numbers confirm these are the same physical drives: two 960 GB M.2 and two 15.36 TB U.2. The move directly tests whether the PCIe link issues seen on .217 follow the drives or the slots.

All four drives passed a NUMA-pinned fio 3.28 suite with zero media errors. The 960 GB drives' sequential read doubled (1.87 -> 3.76 GB/s) and random read nearly doubled (443K -> 852K IOPS) simply by moving to .38 - proving the Gen3 speed cap was a .217 slot/BIOS limitation, now resolved (Gen4). The link still trains at x2 width on both servers, so half-width follows the drives and still leaves ~50% of Gen4 bandwidth unused.

One 15.36 TB drive (S7RKNG0YB01786) was found dead at idle from an APST power-state controller-down fault; an in-place PCIe hot-rescan recovered it to full Gen5 x4 and it completed the entire suite under load. It is electrically sound but needs the APST mitigation before production.

Validation Scorecard

Subsystem	Result	Detail	Verdict
15.36 TB U.2 (Gen5 x4)	12.0 / 12.5 GB/s read, ~1.5M IOPS	On par with .217; full Gen5 x4, 7.05 GB/s write	PASS
960 GB M.2 (Gen4 link)	3.76 GB/s read, 852K IOPS	2x faster than .217 (Gen4 vs Gen3 link fix)	PASS
960 GB M.2 link width	Gen4 x2 (half width)	x4 capable; ~50% of Gen4 bandwidth left on the table	LIMITED
Drive S7RKNG0YB01786	Recovered from APST controller-down	Ran full load OK; needs APST fix before production	ACTION
SMART health (all 4)	0 media errors, 0% wear, <=46 C	No thermal throttling	PASS



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	MDI300

Chassis	2U
Hostname	user-Dev
Serial	n/a

CPU

Model	2x Intel Xeon 6730P
Sockets	2
Cores Per Socket	32
Threads	128
Numa Nodes	2
Numa Device Map	all 4 NVMe slots on node 0 (CPUs 0-31, 64-95)

Memory

Total	1.0 TB
Type	DDR5

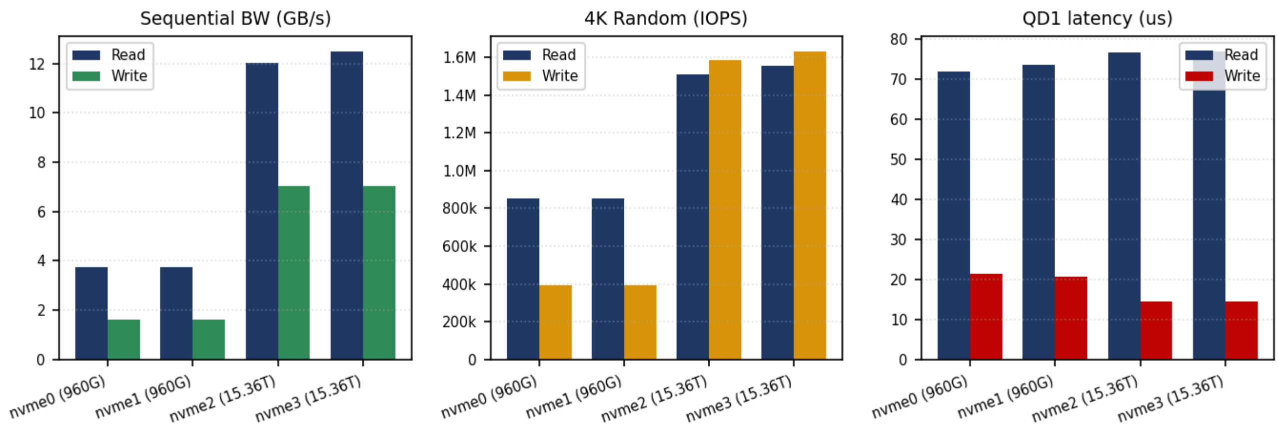
Kernel / OS

Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-111-generic
Cmdline	(recommended) nvme_core.default_ps_max_latency_us=0 pcie_aspm=off pcie_port_pm=off

Storage Devices

Dev	Model	Form	Capacity	Serial	Link (neg)	Link (cap)
/dev/nvme0n1	Samsung PM9D3a	M.2 22x80	960 GB	S8CDNG0YC00174	Gen4 x2	Gen4 x4
/dev/nvme1n1	Samsung PM9D3a	M.2 22x80	960 GB	S8CDNG0YC00153	Gen4 x2	Gen4 x4
/dev/nvme2n1	Samsung PM9D3a	U.2 2.5in	15.36 TB	S7RKNG0YB01786	Gen5 x4 (recovered)	Gen5 x4
/dev/nvme3n1	Samsung PM9D3a	U.2 2.5in	15.36 TB	S7RKNG0YB01763	Gen5 x4	Gen5 x4

Storage Performance



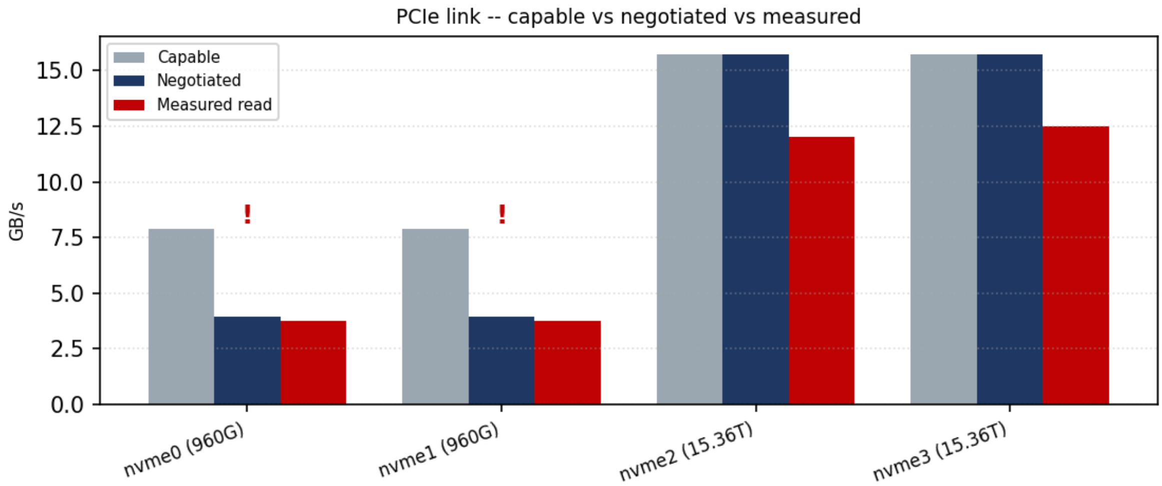
Drive	Link	SeqR	SeqW	RandR	RandW	QD1R	QD1W	p99.9R
nvme0 (960G)	Gen4 x2	3.8	1.6	852k	396k	72	21	n/a

Drive	Link	SeqR	SeqW	RandR	RandW	QD1R	QD1W	p99.9R
nvme1 (960G)	Gen4 x2	3.8	1.6	854k	396k	73	21	n/a
nvme2 (15.36T)	Gen5 x4	12.0	7.0	1.51M	1.59M	77	14	n/a
nvme3 (15.36T)	Gen5 x4	12.5	7.0	1.55M	1.63M	77	14	n/a

Units: SeqR/SeqW in GB/s; RandR/RandW in IOPS; QD1 & p99.9 in microseconds.

PCIe Link Analysis

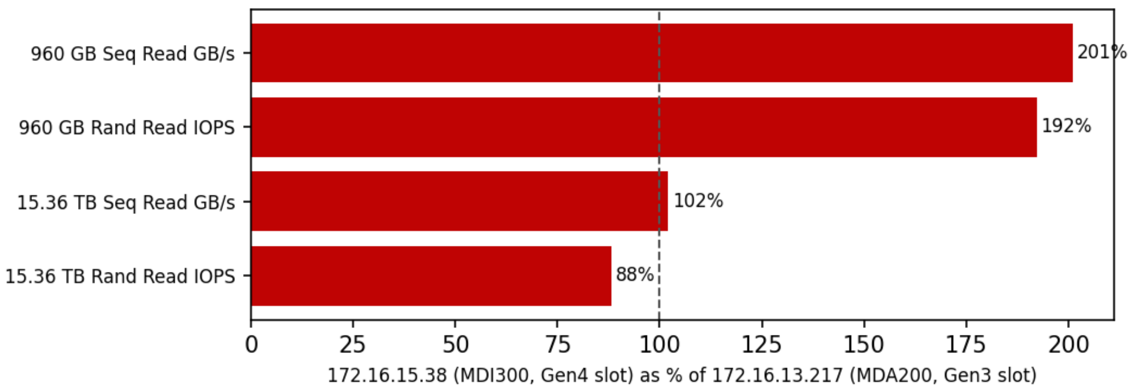
Capable vs negotiated link bandwidth vs measured sequential read. Drives negotiating below their capable width/speed are flagged.



Link warnings: nvme0 (960G): Gen4 x2 negotiated, x4 capable -- ~50% of link bandwidth; nvme1 (960G): Gen4 x2 negotiated, x4 capable -- ~50% of link bandwidth

Cross-Server Comparison

Same four PM9D3a drives across two servers. The 960 GB read scales with link generation (Gen3 -> Gen4); the 15.36 TB Gen5 pair is consistent within run variance.



Metric	172.16.13.217 (MDA200, Gen3 slot)	172.16.15.38 (MDI300, Gen4 slot)	Delta
960 GB Seq Read GB/s	1.87	3.76	+101%
960 GB Rand Read IOPS	443,237	852,199	+92%
15.36 TB Seq Read GB/s	12.22	12.48	+2%

Metric	172.16.13.217 (MDA200, Gen3 slot)	172.16.15.38 (MDI300, Gen4 slot)	Delta
15.36 TB Rand Read IOPS	1,760,240	1,554,033	-12%

960 GB read throughput doubled purely because the MDI300 trains the link at Gen4 (16 GT/s) where the MDA200 was stuck at Gen3 (8 GT/s) -- a .217 platform/slot fault, not the drives. Both servers still negotiate only x2 width, so ~half of a true Gen4 x4 link is unused. 960 GB write is NAND-bound and unchanged by the faster link.

Component Health

Drive	Link	Temp pre->post	Wear	Media errors	Verdict
nvme0n1 (960 GB)	Gen4 x2	35->41 C	0%	0	PASS
nvme1n1 (960 GB)	Gen4 x2	38->46 C	0%	0	PASS
nvme2n1 (15.36 TB, recovered)	Gen5 x4	35->45 C	0%	0	PASS
nvme3n1 (15.36 TB)	Gen5 x4	34->43 C	0%	0	PASS

Kernel log (dmesg) diff: nvme2n1 (S7RKNG0YB01786) reported controller-down at idle (CSTS=0xffffffff, PCI_STATUS=0x10) ~99 min after boot; auto-reset failed (-19) and the device was disabled. Root cause: APST power-state bug (kernel recommended nvme_core.default_ps_max_latency_us=0 / pcie_aspm=off). Recovered in place via PCIe hot remove + bus rescan to full Gen5 x4; passed the full fio suite under load with no further drops.

Findings & Recommendations

Findings

- All four PM9D3a drives are healthy and performant: 0 media errors, 0% wear, clean thermals. The 15.36 TB U.2 pair delivers full Gen5 x4 (~12 GB/s read, 7 GB/s write, ~1.5M IOPS).
- The .217 Gen3 cap was a slot fault, now resolved: the 960 GB drives read 2x faster on .38 (Gen4 vs Gen3). Their write path is NAND-bound (~1.62 GB/s) and unchanged by the faster link.
- The x2 lane-width limit follows the drives: both servers train these M.2 units at x2, leaving ~50% of Gen4 x4 bandwidth unused.
- Drive S7RKNG0YB01786 has an APST idle-stability defect: electrically sound under load, but drops off the bus at idle until power-state transitions are disabled.

Recommendations

- Apply the APST mitigation (priority): boot .38 with nvme_core.default_ps_max_latency_us=0 pcie_aspm=off pcie_port_pm=off (or update PM9D3a firmware), then idle-soak S7RKNG0YB01786 for several hours to confirm it no longer drops. Do not deploy this drive until verified.
- Unlock x4 width for the 960 GB M.2 drives: move them to M.2 slots wired x4 (or check board lane bifurcation); a true Gen4 x4 link should roughly double sequential read again to ~7 GB/s.
- Consider 4K LBA format: all drives are 512-byte; reformat the PM9D3a units to 4096-byte LBA before deployment for best efficiency.
- Monitor in production: enable smartd / nvme-cli alerting for temperature, media errors and controller-down events.

Conclusion

Device	Outcome	Verdict
2x PM9D3a 15.36 TB U.2 (nvme2 + nvme3)	Full Gen5 x4 performance and health	PASS

Device	Outcome	Verdict
nvme2 APST stability (S7RKNG0Y B01786)	Recovered; apply APST mitigation + idle-soak before production	ACTION
2x PM9D3a 960 GB M.2 (nvme0 / nvme1)	Healthy; 2x faster than .217 (Gen4). Still x2 width -- move to an x4 slot	LIMITED

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Fio	fio 3.28
Distro	Ubuntu 22.04.4 LTS
Kernel	Linux 6.8.0-111-generic
Methodology	ioengine=libaio, direct=1 (raw block device), numactl node0 pinning, drives run sequentially
Campaign Window	04 Jun 2026 06:15-06:39 UTC
Recommended Cmdline	nvme_core.default_ps_max_latency_us=0 pcie_aspm=off pcie_port_pm=off
Note	Reconstructed from the validated source report into the CoreBench results.json contract.